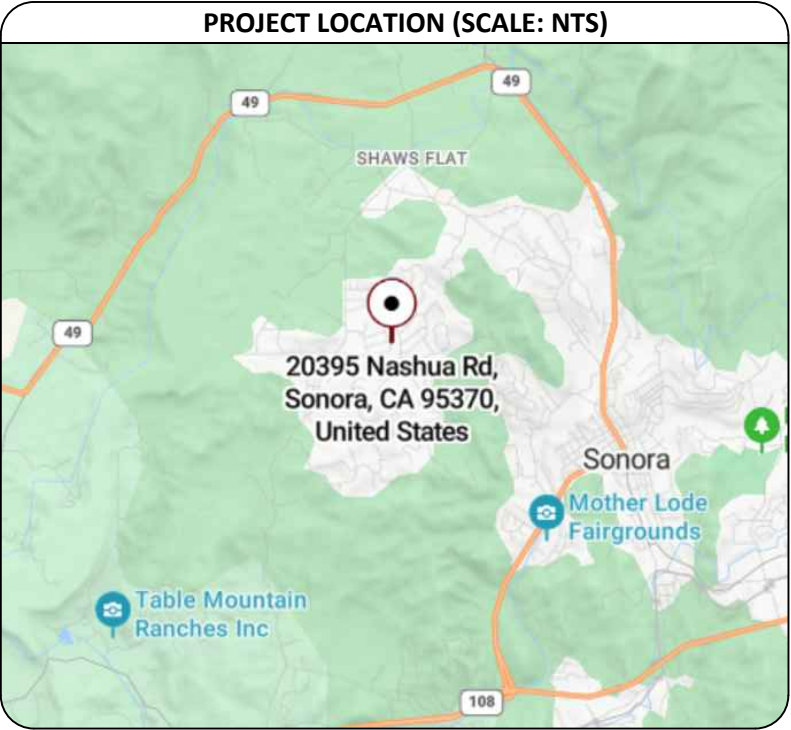
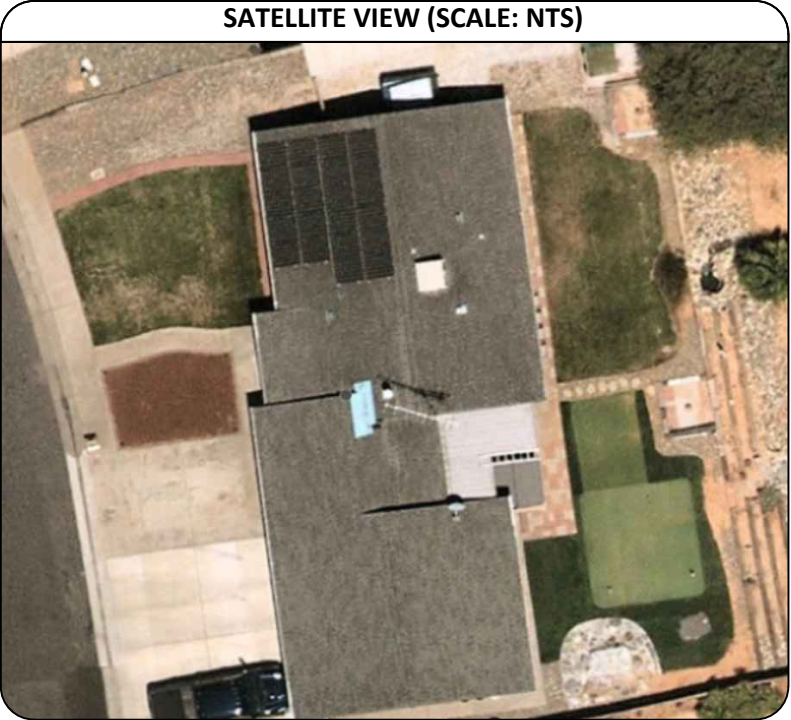


PROJECT: ESS SYSTEM DETAILS	
ESS STORAGE CAPACITY	ESS OUTPUT CAPACITY
(4) BATTERY x 5000WH	(4) BATTERY x 3840W
20.0 kWh	15.3 kW



PROJECT: EXISTING PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(22) MODULES x 145W	(1) INVERTER x 3000W
3.200 kW	3.000 kW

- SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL ESS SYSTEM AT A RESIDENTIAL LOCATION : 3525 STREET NAME, SONORA, CA 68423. COORDINATES: 37.992439,-120.405442
 - THE ESS WILL BE PREASSEMBLED FOR FAST AND EASY INSTALLATION.
 - INSTALLED ESS SYSTEM WILL BE CONNECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ).
 - THE ENERGY STORED BY THE ESS SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES.

- APPLICABLE GOVERNING CODES & NOTES
- THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES:
- 2022 CALIFORNIA BUILDING CODE
 - 2022 CALIFORNIA RESIDENTIAL CODE
 - 2022 CALIFORNIA MECHANICAL CODE
 - 2022 CALIFORNIA PLUMBING CODE
 - 2022 CALIFORNIA ENERGY CODE
 - 2022 CALIFORNIA FIRE CODE
 - 2022 CALIFORNIA ELECTRICAL CODE
- ALL OTHER RELATED STATE, LOCAL & AHJ AMENDMENTS.

- DESIGN CRITERIA
- CONSTRUCTION: SINGLE FAMILY DWELLING (SFD)
 - NO. OF STORIES: SINGLE STORY
 - EXPOSURE CATEGORY : B, RISK CATEGORY : II
 - WIND SPEED (ASCE 7-22) : 94 MPH
 - GROUND SNOW LOAD (ASCE 7-22) : 25 lb/sqft
 - ASHRAE EXTREME LOW: -2°C
 - ASHRAE HIGH TEMP (2% AVG.) : 38°C
 - ASHRAE DISTANCE ABOVE ROOF (7/8") : 60°C

EQUIPMENT DETAILS	
(N) (#) BATTERY	(04) ENPHASE IQ BATTERY 5P (IQBATTERY-5P-1P-NA)
(N) # SMART SWITCH DEVICE	(01) ENPHASE IQ SYSTEM CONTROLLER 3/3G (SC200D111C240US01)
UTILITY	P&GE
AHJ	TUOLUMNE COUNTY

- GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & CEC CERTIFIED, WHERE WARRANTED
 - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS.
 - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER CEC CODE 240.24.
 - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH CEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER CEC
 - GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM.
 - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE.
 - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING. MECHANICAL, OR BUILDING ROOF VENTS.
 - WATER PROOF CONCECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER CEC 312.2 AND 314.15.
 - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER CEC 250.92.
 - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE.
 - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM.
 - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
- PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF DETAILS & BOM
S-1	EQUIPMENT ELEVATION DIAGRAM
E-1	ELECTRICAL DIAGRAM & CALCULATION
E-2	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

- ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (CEC690.4(E) AND 705.6)
 - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONNECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF.
 - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES.
 - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL.
 - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS .
 - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING
 - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES.
 - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT.
 - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO CEC.
 - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED.
 - ALL EXTERIOR CONDUIT TO BE PAINT MATCHED TO THE HOMES COLORS

CONTRACTOR INFO:

DATE	--/------
PROJECT ID	SP-3
CREATED BY	---
CHECKED BY	--
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

ABALOS RESIDENCE
3525 STREET NAME, SONORA, CA 68423
APN: 987456123

#	REVISION/UPDATE	DATE

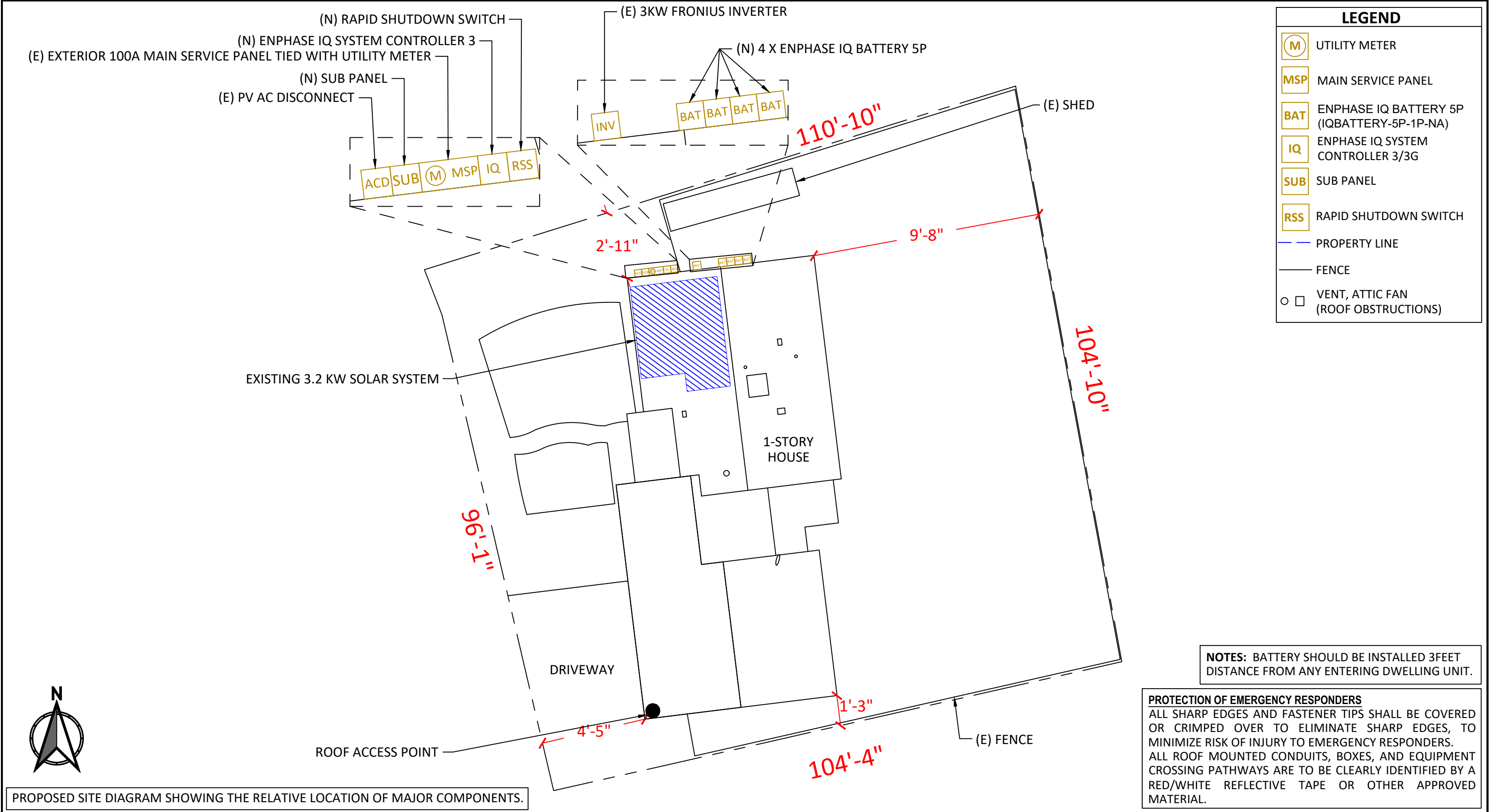
SIGNED & STAMP BY

PV-0

SHEET NO.

PROJECT SUMMARY

SHEET INFORMATION



CONTRACTOR INFO:	DATE	--/------
	PROJECT ID	SP-3
	CREATED BY	---
	CHECKED BY	--
	SCALE	1" = 16'
SHEET SIZE	11"X 17"	
	ANSI B	
HOMEOWNER INFO		

ABALOS RESIDENCE
3525 STREET NAME, SONORA, CA 68423
APN: 987456123

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-1
SHEET NO.

SITE PLAN

SHEET INFORMATION

LEGEND

M

UTILITY METER

MSP

MAIN SERVICE PANEL

BAT

ENPHASE IQ BATTERY 5P (IQBATTERY-5P-1P-NA)

IQ

ENPHASE IQ SYSTEM CONTROLLER 3/3G

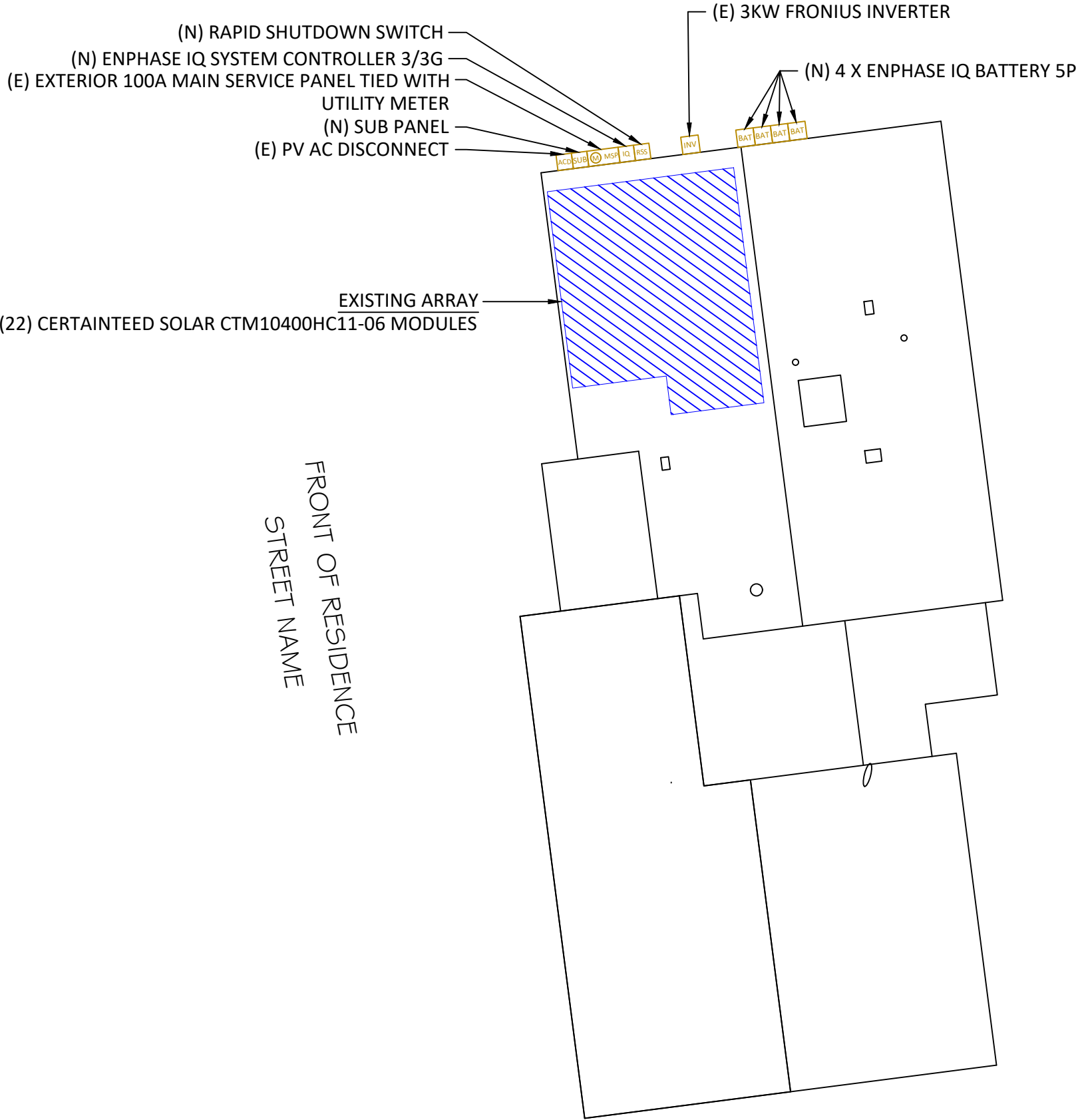
SUB

SUB PANEL

RSS

RAPID SHUTDOWN SWITCH

VENT, ATTIC FAN (ROOF OBSTRUCTIONS)



BILL OF MATERIALS		
EQUIPMENT	QTY	DESCRIPTION
BATTERY	4	ENPHASE IQ BATTERY 5P (IQBATTERY-5P-1P-NA)
SMART PANEL	1	ENPHASE IQ SYSTEM CONTROLLER 3/3G (SC200D111C240US01)
SUB PANEL	1	SUB PANEL, 200A RATING, 240V NEMA 3R
RAPID SHUTDOWN	1	RAPID SHUTDOWN SWITCH

NOTES: BATTERY SHOULD BE INSTALLED 3FEET DISTANCE FROM ANY ENTERING DWELLING UNIT.

CONTRACTOR INFO:

DATE	--/------
PROJECT ID	SP-3
CREATED BY	---
CHECKED BY	--
SCALE	3/32" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

ABALOS RESIDENCE

3525 STREET NAME, SONORA, CA 68423

APN: 987456123

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-2

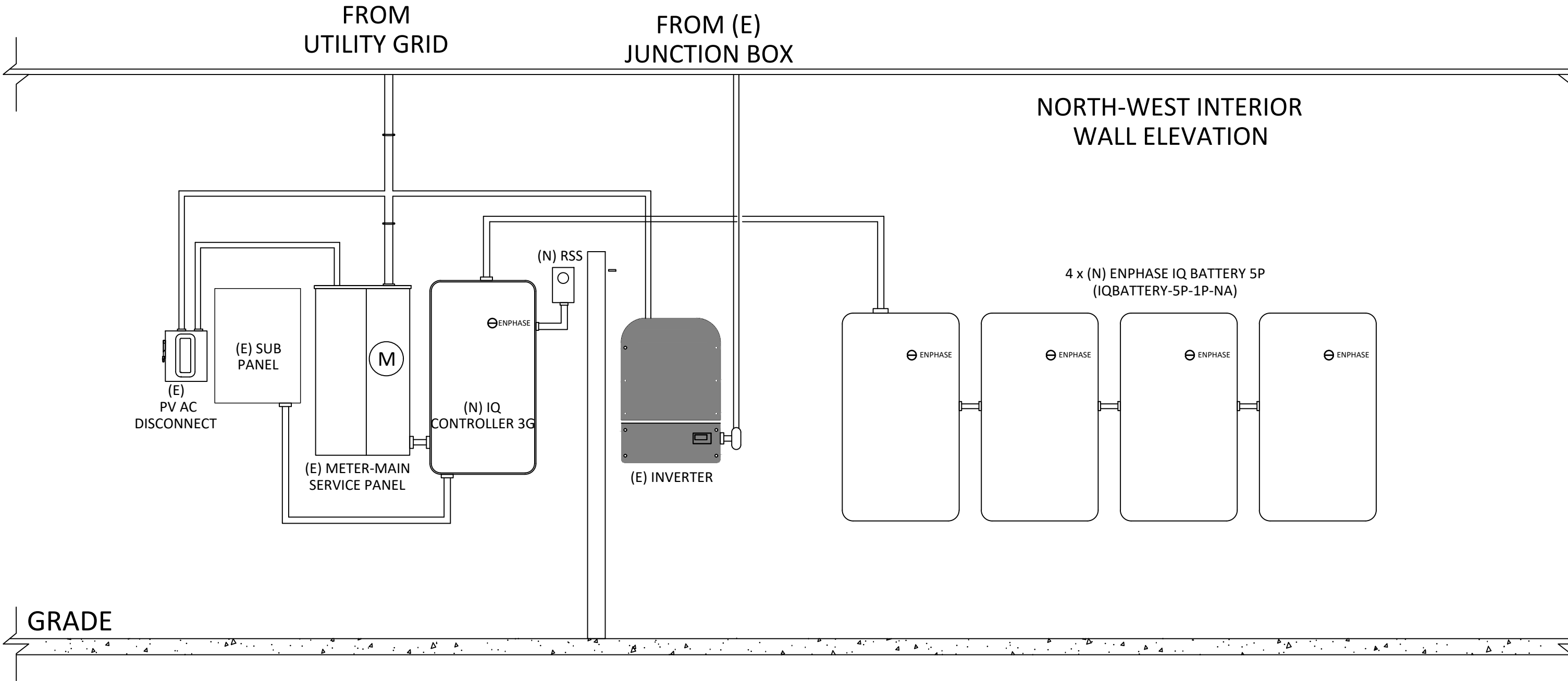
SHEET NO.

ROOF DETAILS & BOM

SHEET INFORMATION

NOTE:
MINIMUM 3' CLEARANCE BETWEEN
BATTERY AND DOOR/WINDOW DIRECTLY
ENTERING THE DWELLING UNIT.

NOTES: INDIVIDUAL ESS UNITS SHALL BE
SEPARATED FROM EACH OTHER BY A MIN. OF 3'
OR PER FIRE DEPT. APPROVED PLANS.



EQUIPMENT ELEVATION DIAGRAM

CONTRACTOR INFO:	
------------------	--

DATE	--/------
PROJECT ID	SP-3
CREATED BY	---
CHECKED BY	--
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO	
----------------	--

ABALOS RESIDENCE
3525 STREET NAME, SONORA, CA 68423
APN: 987456123

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY	
-------------------	--

S-1	
SHEET NO.	
EQUIPMENT ELEVATION DIAGRAM	
SHEET INFORMATION	

EQUIPMENT DETAILS	
(N) (#) BATTERY	(04) ENPHASE IQ BATTERY 5P (IQBATTERY-5P-1P-NA)
(N) # SMART SWITCH DEVICE	(01) ENPHASE IQ SYSTEM CONTROLLER 3/3G (SC200D111C240US01)
UTILITY	P&GE
AHJ	TUOLUMNE COUNTY

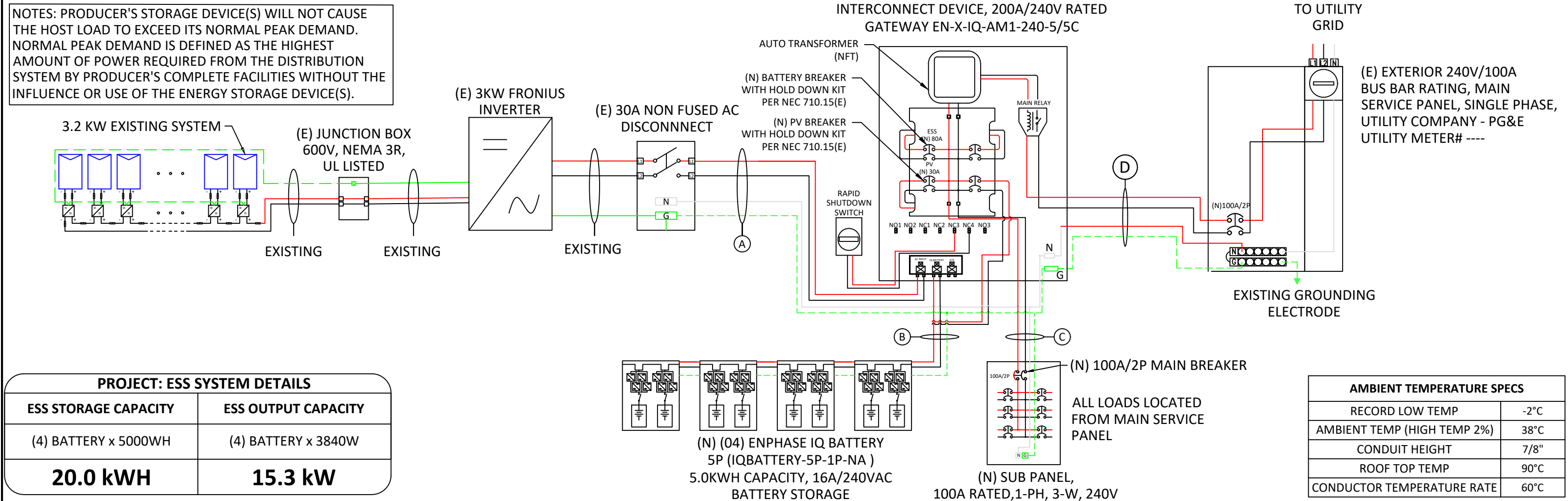
ENPHASE IQ SYSTEM CONTROLLER SPECIFICATION	
MAX. DISCONNECT CURRENT	200A
MAX. OVERCURRENT PROTECTION BREAKER	200A

NOTES: PRODUCER'S STORAGE DEVICE(S) WILL NOT CAUSE THE HOST LOAD TO EXCEED ITS NORMAL PEAK DEMAND. NORMAL PEAK DEMAND IS DEFINED AS THE HIGHEST AMOUNT OF POWER REQUIRED FROM THE DISTRIBUTION SYSTEM BY PRODUCER'S COMPLETE FACILITIES WITHOUT THE INFLUENCE OR USE OF THE ENERGY STORAGE DEVICE(S).

BATTERY SPECIFICATIONS	
MODEL	ENPHASE IQ BATTERY 5P (IQBATTERY-5P-1P-NA)
AMBIENT TEMP (HIGH TEMP 2%)	-20°C TO 50°C
RATED OUTPUT CURRENT	16A
NOMINAL VOLTAGE	240/211-264 VAC
USABLE CAPACITY	5.0 KWH
INTERCONNECTION	SINGLE-PHASE

THE SUMMATION RULE - CEC - 705.12 (B)(3)(3)
NOTES: THIS EQUIPMENT FED BY MULTIPLE SOURCES. TOTAL RATING OF ALL OVERCURRENT DEVICES, EXCLUDING MAIN SUPPLY OVERCURRENT DEVICES SHALL NOT EXCEED AMPACITY OF BUSBAR.
NOTES: BATTERY SHOULD BE INSTALLED 3FEET DISTANCE FROM ANY ENTERING DWELLING UNIT.

PROJECT: EXISTING PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(22) MODULES x 145W	(1) INVERTER x 3000W
3.200 kW	3.000 kW



PROJECT: ESS SYSTEM DETAILS	
ESS STORAGE CAPACITY	ESS OUTPUT CAPACITY
(4) BATTERY x 5000WH	(4) BATTERY x 3840W
20.0 kWH	15.3 kW

AMBIENT TEMPERATURE SPECS	
RECORD LOW TEMP	-2°C
AMBIENT TEMP (HIGH TEMP 2%)	38°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	90°C
CONDUCTOR TEMPERATURE RATE	60°C

ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT (IQ8PLUS CONTINUOUS CURRENT x NO. OF IQ8PLUS)				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	10 AWG	THWN	COPPER	MIN 0.75" Dia EMT	1	3	18.13%	30A	8 AWG	THWN, COPPER	0.88	(38°C)	1	24.0A	x	1.25	=	30.0A	35A	35A x 1 x 0.88 = 30.8A	5FT	0.13%
B	4 AWG	THWN	COPPER	MIN 0.75" Dia EMT	1	2	36.47%	80A	8 AWG	THWN-2, COPPER	0.88	(38°C)	1	80.0A	x	1.25	=	100A	85A	85A x 1 x0.88 = 74.8A	15FT	0.29%
C	1 AWG	THWN	ALUMINIUM	MIN 1.25" Dia EMT	1	3	29.25%	100A	8 AWG	THWN-2, COPPER	0.88	(38°C)	1	80.0A	x	1.25	=	100A	100A	100A x 1 x0.88 = 88A	5FT	0.05%
D	1 AWG	THWN	ALUMINIUM	MIN 1.25" Dia EMT	1	3	29.25%	100A	8 AWG	THWN-2, COPPER	0.88	(38°C)	1	80.0A	x	1.25	=	100A	100A	100A x 1 x0.88 = 88A	5FT	0.05%

	DATE	--/------			
	PROJECT ID	SP-3			
	CREATED BY	---			
	CHECKED BY	--			
	SCALE	NTS			
	SHEET SIZE	11"X 17"			
ANSI B					
CONTRACTOR INFO:		ABALOS RESIDENCE 3525 STREET NAME, SONORA, CA 68423 APN: 987456123	#	REVISION/UPDATE	DATE
		HOMEOWNER INFO:	SIGNED & STAMP BY		
			E-1		
			SHEET NO.		
			ELECTRICAL DIAGRAM & CALCULATIONS		
			SHEET INFORMATION		

1

CAUTION - TRI POWER SOURCE

FIRST SOURCE IS UTILITY ELECTRICAL GRID
SECOND SOURCE IS AC BATTERY
THIRD SOURCE IS PV INVERTER

LABEL LOCATION:
SUB PANEL OR BACKUP LOADS PANEL, BATTERY, INVERTER OUTPUT
PANEL, MAIN SERVICE PANEL, METER/MAIN AND PV AC DISCONNECT
2022 CEC 705.10

2

WARNING

THIS EQUIPMENT FED BY MULTIPLE SOURCES.
TOTAL RATING OF ALL OVERCURRENT DEVICES,
EXCLUDING MAIN SUPPLY OVERCURRENT DEVICES
SHALL NOT EXCEED AMPACITY OF BUSBAR.

REQ'D BY: CEC 705.12(B)(3)(3)

3

BATTERY DISCONNECT

DO NOT RELOCATE
THIS OVERCURRENT
DEVICE

REQ'D BY: CEC 706.15(C)
LABEL LOCATION:
BATTERY BREAKERS

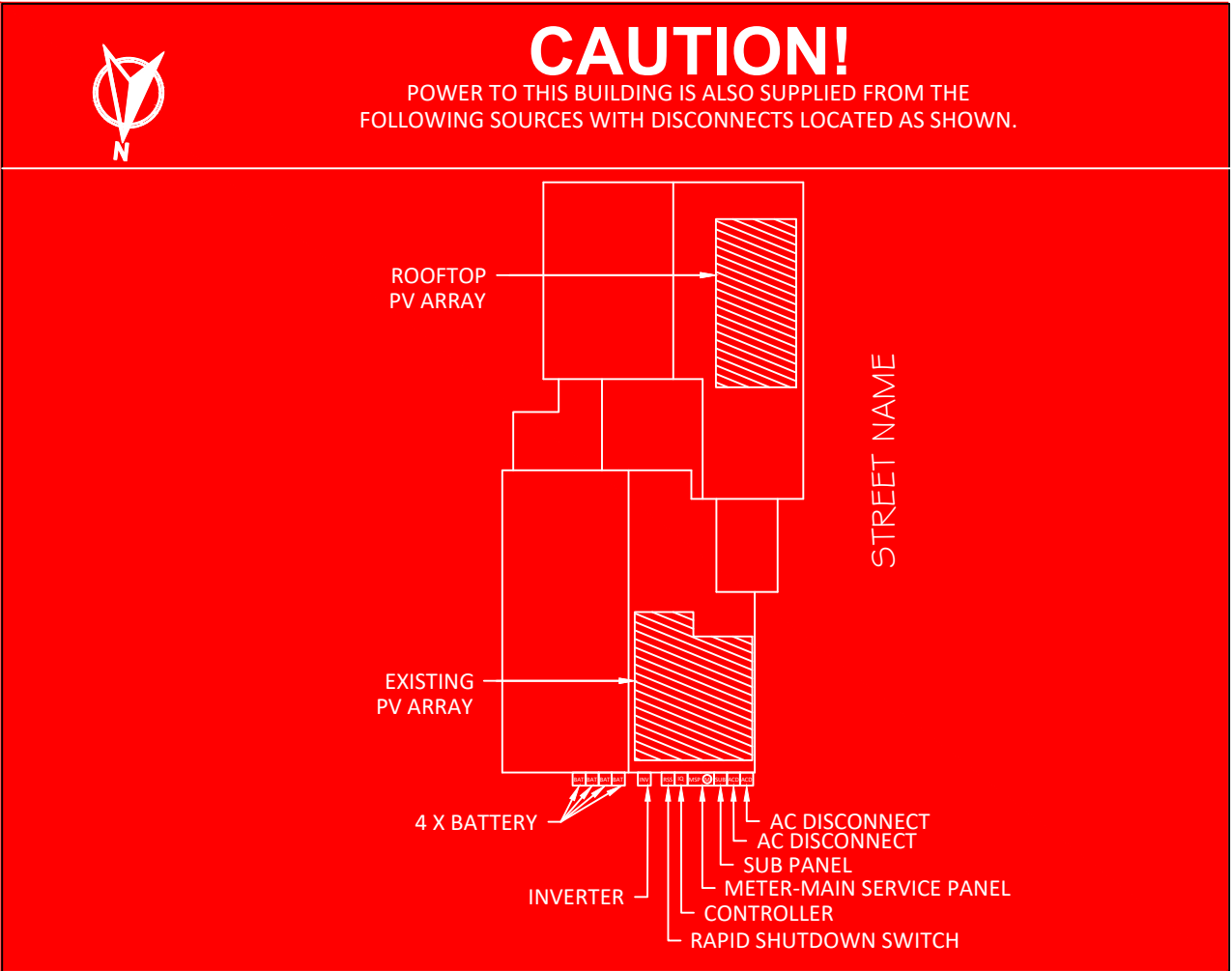
4

PV SOLAR BREAKER

DO NOT RELOCATE THIS
OVERCURRENT DEVICE

LABEL LOCATION:
MAIN SERVICE PANEL
2022 CEC 705.12(B)(3)(2)

ADHESIVE FASTENED SIGNS:
• LABELS SHALL BE MADE OF RED PLASTIC MATERIAL WITH ENGRAVED WHITE LETTERS.
• LETTERS SHALL BE A MINIMUM 3/8" IN SIZE.
• THE LABELS SHALL BE PERMANENTLY ATTACHED TO THE APPROPRIATE PANEL.
• AC & DC CONDUIT, RACEWAY, ENCLOSURES, CABLE ASSEMBLIES AND JUNCTION BOXES
SHALL BE RED BACKGROUND MATERIAL WITH WHITE LETTERING MADE OF DURABLE
ADHESIVE, REFLECTIVE, WEATHER RESISTANT MATERIAL SUITABLE FOR THE
ENVIRONMENT PER UL 969; TO ALERT FIRE SERVICE TO AVOID CUTTING THEM OFF.



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

CONTRACTOR INFO:

DATE	--/------
PROJECT ID	SP-3
CREATED BY	---
CHECKED BY	--
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

ABALOS RESIDENCE
3525 STREET NAME, SONORA, CA 68423
APN: 987456123

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-2
SHEET NO.

PLACARDS

SHEET INFORMATION



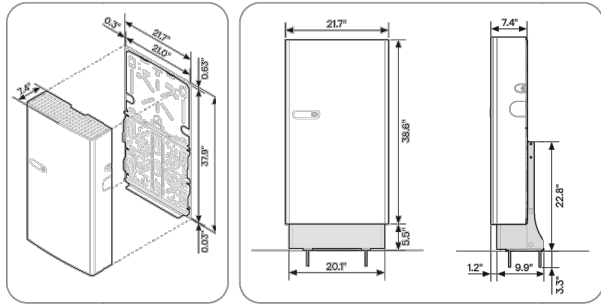
DATASHEET NA



IQ Battery 5P

The IQ Battery 5P all-in-one AC-coupled system is powerful, reliable, simple, and safe. It has a total usable energy capacity of 5.0 kWh and includes six embedded grid-forming microinverters with a 3.84 kVA continuous power rating. It provides backup capability, and installers can quickly design the right system size to meet the customer needs.

Dimensions in inches



Wall mounted

Floor mounted with pedestal
(sold separately)



LISTED

UL 9540A
Certified

Powerful

- Provides 3.84 kVA continuous and 7.68 kVA peak power
- Doubles the available power per kWh of prior generations of IQ Battery
- Includes six embedded IQ8D-BAT Microinverters

Reliable

- 15-year limited warranty
- Cools passively with no moving parts or fans
- Uses wired communication for fast and consistent connection
- Updates software and firmware remotely

Simple

- Fully integrated AC battery system
- Installs and commissions easily
- Supports Backup, Self-Consumption, and time-of-use (TOU) modes
- Offers homeowners remote monitoring and control from the Enphase App
- Field replaceable components

Safe

- Evaluated to UL 9540A for large scale fire testing and reduced separation distance as required in 2021 IRC R328.3.1, 2021 IFC 1207.1.5, and 2023 NFPA 855 15.3.1 and 9.1.5.¹
- Uses lithium iron phosphate (LFP) chemistry for maximum safety and longevity

¹Follow all installation instructions and local codes and requirements of the Authority Having Jurisdiction (AHJ) when installing Enphase ESS.

IQ Battery 5P

MODEL NUMBER	
IQBATTERY-5P-1P-NA	The IQ Battery 5P system with integrated IQ Microinverters and battery management system (BMS) with battery controller
WHAT'S IN THE BOX	
IQ Battery 5P unit	IQ Battery 5P unit (B05-T02-US00-1-3)
ID cover and conduit cover	IQ Battery 5P cover with two conduit covers for the left and right sides of the unit
Bottom mounting bracket and top shield	Bottom mounting bracket for mounting the battery on the wall. One top shield is required for UL9540A
M5 seismic screws	Two M5 seismic screws for securing the battery unit on the bottom mounting bracket
M4 grounding screws	Two M4 grounding screws for securing the top shield on the bottom mounting bracket
M5 ID cover grounding screws	Two M5 ID cover grounding screws for the EMI/EMC requirement
Cable ties	Six cable ties for securing field cables to the unit
Control (CTRL) connector	Spare CTRL connector without resistor for CTRL wiring
Control (CTRL) connector with resistor	Spare CTRL connector with resistor for CTRL wiring
Quick Install Guide (QIG)	QIG for IQ Battery unit installation instructions
OPTIONAL ACCESSORIES AND REPLACEMENT PARTS	
IQ8D-BAT-RMA	IQ8D-BAT Microinverter for field replacement
B05-T02-US00-1-3-RMA	IQ Battery 5P Battery unit for field replacement
B05-CX-0550-O	IQ Battery 5P cover for field replacement
B05-PI-0550-O	IQ Battery 5P pedestal mount
B05-CP-096-O	IQ Battery 5P conduit plates for field replacement. Includes one left-side and one right-side conduit plate
B05-WB-0543-O	IQ Battery 5P wall bracket for field replacement. Includes one bottom mounting bracket and one top shield
IQBATTERY-HNDL-5	IQ Battery 5P lifting handles. Includes one left-side and one right-side lifting handle
B05-ACFB-080-O	IQ Battery 5P AC filter board for field replacement
B05-BMSNA-0490-O	IQ Battery 5P BMS board for field replacement
B05-CANB-063-O	IQ Battery 5P control communication board for field replacement
B05-NICS-0524-O, B05-NUCS-0524-O	IQ Battery 5P control switch is preinstalled on the wiring cover for field replacement
OUTPUT (AC)	
Rated (continuous) output power	3.84 kVA
Peak output power	7.68 kVA (3 seconds), 6.14 kVA (10 seconds)
Nominal voltage/range	240/211–264 VAC
Nominal frequency/range	60/57–63 Hz
Rated output current (@240 VAC)	16 A
Peak output current (@240 VAC)	32 A (3 seconds), 25.6 A (10 seconds)
Load start capability	Up to 48 A LRA ²
Power factor (adjustable)	0.85 leading...0.85 lagging
Maximum units per 20 A branch circuit	One unit (single-phase)
Maximum conductor size supported	3 AWG
Overcurrent protection device (OCPD) for 3 AWG cable	80 A
Interconnection	Single-phase
AC round-trip efficiency ⁴	90%

²Supported in both grid-connected and backup/off-grid operation

³Load start capability may vary

⁴AC to the battery to AC at 50% power rating

IQ Battery 5P

BATTERY	
Total capacity	5.0 kWh
Usable capacity	5.0 kWh
DC round-trip efficiency	96%
Nominal DC voltage	76.8 V
Maximum DC voltage	86.4 V
Ambient operating temperature range (charging)	-20°C to 50°C (-4°F to 122°F) non-condensing
Ambient operating temperature range (discharging)	-20°C to 55°C (-4°F to 131°F) non-condensing
Optimum operating temperature range	0°C to 30°C (32°F to 86°F)
Chemistry	Lithium iron phosphate (LFP)
MECHANICAL DATA	
Dimensions (HxWxD)	980 mm x 550 mm x 188 mm (38.6 in x 21.7 in x 7.4 in)
Lifting weight	66.3 kg (146.1 lbs)
Total installed weight	78.9 kg (174 lbs)
Enclosure	Outdoor–NEMA 3R
IQ8D-BAT Microinverter enclosure	NEMA type 6
Cooling	Natural convection
Altitude	Up to 2,500 meters (8,202 feet)
Mounting	Wall-mount or pedestal-mount (sold separately)
FEATURES AND COMPLIANCE	
Compatibility	Compatible with IQ and M Series Microinverters, IQ System Controller 3/3G, IQ Combiner 5/5C, and IQ Gateway for grid-tied and backup operation
Communication	Wired control communication
Services	Backup, Self-Consumption, TOU, and NEM integrity
Monitoring	Enphase Installer Platform and Enphase App monitoring options; API integration
Compliance	CA Rule 21 (UL 1741-SA), IEEE 1547:2018 (UL 1741-SB, 3rd Ed.) CAN/CSA C22.2 No. 1073-16 UL 9540 ³ , UL 9540A, UN 38.3, UL 1998, UL 991, NEMA Type 3R, AC156 EMC: 47 CFR, Part 15, Class B, ICES 003 Cell module: UL 1973, UN 38.3 Inverters: UL 62109-1, IEC 62109-2
LIMITED WARRANTY	
Limited warranty	>60% capacity, up to 15 years or 6,000 cycles ⁶

³ Following local standards, choose a well-ventilated, non-habitable, indoor location (like a 2-car garage) or in an outdoor location, which is out of direct sunlight and where the ambient temperature and humidity are within -4°F to 113° F (-20°C to 45°C) and 5% to 95% RH, non-condensing.

⁶ Whichever occurs first. Restrictions apply



IQ System Controller 3/3G

The Enphase IQ System Controller 3/3G connects the home to grid power, the IQ Battery system, and solar PV. It provides microgrid interconnect device (MID) functionality by automatically detecting and seamlessly transitioning the home energy system from grid power to backup power in the event of a grid failure. It consolidates interconnection equipment into a single enclosure and streamlines grid-independent capabilities of PV and storage installations by providing a consistent, pre-wired solution for residential applications.



IQ Series Microinverters

The high-powered smart grid-ready IQ Series Microinverters (M Series, IQ6, IQ7, and IQ8 Series) dramatically simplify the installation process



IQ Combiner 5/5C

Consolidates PV interconnection equipment into a single enclosure and streamlines IQ Series Microinverters and IQ Gateway installation by providing a consistent, pre-wired solution for residential applications



10-year limited warranty



IQ Battery 5P

Fully integrated AC battery system. Includes six field-replaceable IQ8D-BAT microinverters



IQ Load Controller

Helps prioritize essential appliances during a grid outage to optimize energy consumption and prolong battery life

Easy to install

- Connects to service entrance¹ or main load center
- Includes neutral-forming transformer
- Mounts on single stud with centered brackets
- Provides conduit entry from the bottom, left, or right
- Includes color-coded wires for ease of wiring the System Shutdown Switch
- Integrates hold-down functionality to eliminate the need for hold-down kits and special breakers

Flexible

- Can be used for Sunlight Backup, Home Essentials Backup, or Full Energy Independence
- IQ System Controller 3 integrates with IQ Battery 5P
- IQ System Controller 3G integrates with select AC standby generators. See the [Generator integration tech brief](#) for a list of generators
- Provides a seamless transition to backup

Safe and reliable

- System Shutdown Switch can be used to disconnect PV, battery, and generator systems
- System Shutdown Switch acts as a rapid shutdown initiator of grid-forming IQ8 PV Microinverters for the safety of maintenance technicians/first responders
- 10-year limited warranty

IQ System Controller 3/3G

MODEL NUMBER	DESCRIPTION
SC200D111C240US01	IQ System Controller 3 streamlines the grid-independent capabilities of PV and storage installations. Integrates hold-down capability. Supports IQ Battery 5P units up to 40 kWh (without PCS*) and 80 kWh (with PCS*). Does not support generator integration
SC200G111C240US01	IQ System Controller 3G streamlines the grid-independent capabilities of PV and storage installations. Integrates hold-down capability. Supports IQ Battery 5P units up to 20 kWh (without PCS*) and 40 kWh (with PCS*). Supports generator integration
WHAT'S IN THE BOX	
IQ System Controller 3/3G	Includes neutral-forming transformer (NFT) and microgrid interconnect device (MID)
System Shutdown Switch	Includes pre-wired red, black, orange, and purple 12 AWG wire (EP200G-NA-02-RSD)
Wall-mounting bracket	Screws provided in the accessories kit for mounting
4-pole circuit breaker	Pre installed Quad breaker (BRK-20A40A-4P-240V), 20 A-40 A, 10 kAIC, Eaton BQC220240 ²
Accessories kit	IQ System Controller 3/3G literature kit, including labels, CTRL headers, screws, filler plates, and Quick Install Guide (QIG) (EP200G-LITKIT)
OPTIONAL ACCESSORIES AND REPLACEMENT PARTS	
CT-200-SPLIT	200 A split core current transformers for metering (accuracy: ±2.5%) ³
CT-200-CLAMP	200 A clamp-type current transformers for metering (accuracy: ±2.5%) ³
Main or load circuit breakers (order separately, as needed) ⁴	• BRK-100A-2P-240V : 2-pole, 100A, 25kAIC, CSR2100N or CSR2100 • BRK-125A-2P-240V: 2-pole, 125A, 25kAIC, CSR2125N • BRK-150A-2P-240V: 2-pole, 150A, 25kAIC, CSR2150N • BRK-175A-2P-240V: 2-pole, 175A, 25kAIC, CSR2175N • BRK-200A-2P-240V: 2-pole, 200A, 25kAIC, CSR2200N
Distributed energy resource (DER) circuit breakers (order separately, as needed) ⁵	• BRK-20A-2P-240V-B: 2-pole, 20 A, 10 kAIC, BR220B/BR220 • BRK-30A-2P-240V-B: 2-pole, 30 A, 10 kAIC, BR230 • BRK-40A-2P-240V-B: 2-pole, 40 A, 10 kAIC, BR240B/BR240 • BRK-60A-2P-240V: 2-pole, 60 A, 10 kAIC, BR260 • BRK-80A-2P-240V: 2-pole, 80 A, 10 kAIC, BR280
EP200G-HNDL-R1	IQ System Controller 3/3G installation handle kit (order separately)
CTRL-SC3-NA-01	Control cable, 500 ft. spool (order separately)
ALTERNATE DER CIRCUIT BREAKERS	
GE/ABB	THQL21xx (20/40/60/80 A)
Siemens	Q2xx (20/40/60/80 A)
Siemens (quad breaker)	Q24020CT2 (20/40 A)
ELECTRICAL SPECIFICATIONS	
Nominal voltage/Range (L-L)	240 V~ ⁶ ±20%
Voltage measurement accuracy	±1% V nominal (±1.2V L-N and ±2.4V L-L)
Auxiliary (dry) contact for load control, excess PV control, and generator two-wire control	24 V, 1 A
Nominal frequency/Range	60 Hz/56–63 Hz
Frequency measurement accuracy	±0.1 Hz
Maximum continuous current rating	160 A
Maximum input overcurrent protection device	200 A
Maximum output overcurrent protection device	200 A
Maximum overcurrent protection device rating for generator circuit	80 A (IQ System Controller 3G only - SC200G111C240US01)
Maximum overcurrent protection device rating for storage circuit	2 x 80 A (IQ System Controller 3 - SC200D111C240US01), 1 x 80 A (IQ System Controller 3G - SC200G111C240US01)

(2) Factory installed quad breaker (Siemens or Eaton). NFT pre-wired to 40 A terminal of the quad breaker.

(3) Two units of CT-200-SPLIT or CT-200-CLAMP must be bought separately for generator integration.

(4) The IQ System Controller 3 is rated at 22 kAIC.

(5) Integrated hold-down kit support breakers (BR230/BR230/BR240) without predrilled hole.

Integrated hold-down kit also supports GE/ABB and Siemens as mentioned under section alternate DER circuit breakers.

(6) "~" indicates alternating current (AC) supply.

(*) Power Control System.

(1) IQ System Controller 3 is not suitable for use as service equipment in Canada.